Benchmarking Large Vision-Language Models on Fine-Grained Image Tasks: A Comprehensive Evaluation

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Abstract

Recent advancements in Large Vision-Language Models (LVLMs) have demonstrated remarkable multimodal perception capabilities, garnering significant attention. While numerous evaluation studies have emerged, assessing LVLMs both holistically and on specialized tasks, fine-grained image tasks—fundamental to computer vision—remain largely unexplored. To fill this gap, we introduce a comprehensive fine-grained evaluation benchmark, *i.e.*, `FG-BMK`, comprising 1.01 million questions and 0.28 million images. Our evaluation systematically examines LVLMs from both human-oriented and machine-oriented perspectives, focusing on their semantic recognition and fine-grained feature representation capabilities. Through extensive experiments on twelve representative LVLMs/VLMs, we uncover key findings regarding the influence of training paradigms, modality alignment, perturbation susceptibility, and fine-grained category reasoning on task performance. This work provides critical insights into the limitations of current LVLMs and offers guidance for future data construction and model design in the development of more advanced LVLMs. Our code is open-source and available at `https://anonymous.4open.science/r/FG-BMK-7B51`.

1 Introduction

Large language models have recently achieved remarkable advancements, with models like GPT-4 [1] surpassing human-level performance across diverse tasks. Building on this, Large Vision-Language Models (LVLMs) have rapidly evolved, with models such as GPT-4o [1], Qwen [2], InternVL [8], and LLaVA-1.5 [24] demonstrating strong multimodal reasoning and perception capabilities.

In response to these advancements, various holistic and specialized evaluations have emerged to assess the capabilities of LVLMs. For instance, LVLM-eHub [48] and MMBench [51] provide broad evaluations of multimodal perception and reasoning, while specialized evaluations like DocVQA [31] and GQA [15] focus on specific tasks, such as document visual perception and visual reasoning. While these evaluations have provided valuable insights, a critical gap remains: there is no systematic evaluation of LVLMs on fine-grained image tasks, which focus on analyzing visual objects at the subordinate category level and are fundamental to computer vision [44]. Consequently, the capability boundaries of LVLMs in fine-grained tasks remain poorly understood.

To address this gap, we propose a comprehensive fine-grained evaluation benchmark, termed `FG-BMK`. This benchmark includes 1.01 million questions and 0.28 million images, providing a robust test bed for evaluating LVLMs. `FG-BMK` incorporates two evaluation paradigms: human-oriented and

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machine-oriented. The human-oriented evaluation employs dialogue-like interactions to assess a model's ability to understand and respond to fine-grained visual queries in a conversational context. The machine-oriented evaluation focuses on two core fine-grained vision tasks—image retrieval and recognition—to directly measure the feature representation capabilities of LVLMs. Together, these evaluations enable a comprehensive assessment of LVLMs' fine-grained feature representation and semantic recognition abilities.

Through extensive evaluations across representative LVLMs and VLMs within FG-BMK, we derive several key insights:

- The contrastive training paradigm in LVLMs proves more effective in enhancing the fine-grained discriminability of visual features, whereas generative and reconstruction-based training paradigms tend to yield weaker discriminability.
- Aligning visual features with textual features in LVLMs can impair their fine-grained discriminability, particularly when there is a mismatch in granularity between the paired image-text data.
- LVLMs and LVMs are more vulnerable to feature perturbations in fine-grained tasks than in generic vision tasks.
- LVLMs demonstrate relatively stronger capabilities in perceiving visual appearances but face challenges in fine-grained category reasoning (which depends on the recognition of visual attributes).
- Despite their advancements, LVLMs still lag behind fine-grained tailored models in handling fine-grained visual tasks.

These findings provide a foundation for future research aimed at advancing LVLM performance in fine-grained tasks and addressing the challenges uncovered by FG-BMK.

2 Related Work

Large Vision-Language Models Large Language Models (LLMs) like GPT-4 [1] have achieved significant advancements in text comprehension, reasoning, and generation. Building on this foundation, Large Vision-Language Models (LVLMs) have emerged, demonstrating remarkable reasoning and perception capabilities across various tasks. Efforts to enhance LVLM performance have taken diverse approaches. BLIP [22] leverages noisy web data with bootstrapped captions, while LLaVA [24] utilizes GPT-4-generated instruction-following data to improve visual-linguistic understanding. BLIP-2 [21] focuses on integrating frozen image and text encoders, and Qwen2.5-VL [2] employs dynamic-resolution strategy to preserve native image resolution. InternVL3 [55] adopts a unified pre-training approach over multimodal data to improve effectiveness. Despite their success across various tasks, these LVLMs' capabilities in fine-grained visual tasks remain underexplored. Our work seeks to fill this gap by evaluating their performance in this challenging domain, providing deeper insights into their potential and limitations.

Large Vision-Language Model Benchmarks While LVLMs have shown impressive performance, various benchmarks have been developed to evaluate their capabilities in different domains. Specialized benchmarks like ChartQA [30] for chart understanding, DocVQA [31] for document visual perception, and GQA [15] for visual reasoning provide focused assessments. Additionally, evaluations in optical character recognition [25] and adversarial robustness [28] offer insights into specialized aspects of LVLM performance. Holistic evaluations, such as LVLM-eHub [48] and MM-Bench [51], assess general multimodal perception and reasoning, while others like MathVista [27] and MMMU [52] include expert-level problems spanning multiple disciplines.

However, comprehensive evaluations of LVLMs in fine-grained visual tasks remain scarce. Existing efforts, such as [12], primarily focus on fine-grained classification, and [54] evaluates classification and reasoning with limited questions. To address it, we propose FG-BMK, a benchmark designed to comprehensively evaluate LVLMs' fine-grained feature representation and semantic recognition capabilities across diverse visual tasks.

Fine-Grained Image Tasks Fine-grained visual tasks [44, 49, 45, 47, 16, 53] are pivotal in applications like biodiversity monitoring [17], object retrieval [35], and product recommendation [43]. While LVLMs such as GPT-4, InternVL, and Qwen excel in tasks like OCR, visual question answering, and image captioning, fine-grained tasks remain a significant challenge. These tasks demand

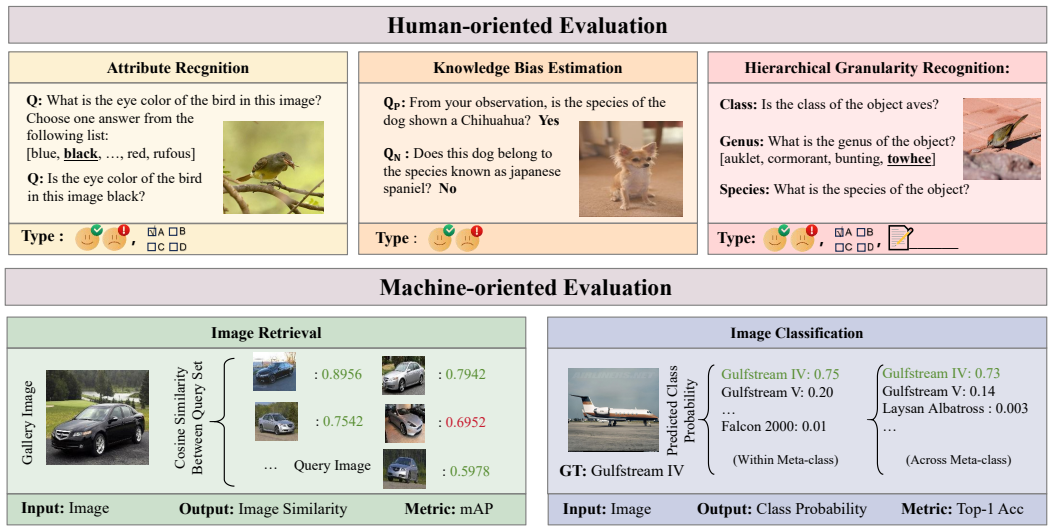

Figure 1: Our proposed benchmark: The human-oriented evaluation tests the model's ability to handle fine-grained visual queries (true/false, multiple-choice, short-answer), while the machine-oriented evaluation directly assesses visual feature representation through image retrieval and classification tasks. =true/false question, =multiple-choice question, =short-answer question.

models to identify subtle, discriminative patterns in images [37] and leverage expert knowledge from LLMs for precise responses. To address this, we introduce a comprehensive benchmark and conduct extensive experiments to evaluate LVLMs' performance on fine-grained tasks. Our study systematically highlights their limitations and offers actionable insights for advancing model design and training.

3 The Evaluation Benchmark

3.1 Evaluation Tasks and Metrics

To comprehensively evaluate LVLMs, we adopt two evaluation paradigms: 1) *human-oriented evaluation* and 2) *machine-oriented evaluation*.

Specifically, since dialogue is the most direct way for humans to interact with LVLMs, the human-oriented evaluation assesses the model's ability to understand and respond to fine-grained visual queries in a conversational setting. We utilize three types of questions—true/false, multiple-choice, and short-answer (cf. Figure 1)—where a response is considered correct if it includes the ground truth.

While, the machine-oriented evaluation encompasses two fundamental vision tasks [44]: image retrieval and image classification. To evaluate the feature representation ability of LVLMs, we measure the accuracy of their visual features on these tasks. Following the DINOv2 approach [33], we use mean Average Precision (mAP) for retrieval and Top-1 accuracy for classification.

The detailed aspects of both human-oriented and machine-oriented evaluations are outlined below:

Human-oriented Evaluation:

- *Attribute Recognition*: This includes true/false questions and multiple-choice questions, designed to evaluate the model's ability to identify visual attributes such as size, color, length, shape, pattern, and more.
- *Knowledge Bias Estimation*: This section consists exclusively of true/false questions, designed to uncover potential knowledge biases across different fine-grained categories by evaluating LVLM's accuracy for each category.
- *Hierarchical Granularity Recognition*: This section includes true/false questions, multiple-choice questions, and short-answer questions, designed to evaluate the LVLMs' ability to leverage domain-

specific knowledge to identify object categories in images across different levels of granularity within hierarchical taxonomies.

Machine-oriented Evaluation:

- *Image Retrieval*: retrieves images belonging to multiple subordinate categories of a meta-category by measuring the similarity of their visual features.
- *Image Classification*: recognizes images into fine-grained categories, either within a single meta-category (*e.g.*, species of animals, models of cars) or across multiple meta-categories, assessing the model's ability to handle diverse data sources and make accurate predictions.

More evaluation tasks details are presented in Appendix A.1.

3.2 Data Curation

To ensure the quality and diversity of the data, we source images for the `FG-BMK` benchmark from 12 well-established fine-grained datasets, which helps avoid issues of inconsistent image quality and the risk of mislabeling often found in web-sourced images [34].

- *Attribute Recognition*: We design true/false and multiple-choice questions based on the fine-grained annotations of the images. For the multiple-choice questions, the options include all possible attribute candidates. In contrast, for the true/false questions, half are negative samples, pairing the image with an incorrect attribute, and the other half are positive samples.
- *Knowledge Bias Estimation*: In addition to positive samples, for each fine-grained category label, we pair it with images from other subcategories within the same super-category to generate negative samples for true/false questions.
- *Hierarchical Granularity Recognition*: We design true/false, multiple-choice, and short-answer questions at various levels of granularity based on the hierarchical taxonomy labels of the images. For true/false questions, negative samples are created by pairing images with incorrect labels at the same hierarchical level (*e.g.*, an image of Aves (birds) paired with Insecta (insects)). For multiple-choice questions, options are drawn from different categories within the same parent category of the hierarchical taxonomy (*e.g.*, species-level options such as black-footed albatross and Laysan albatross, both within the genus Albatross). For short-answer questions, the model is prompted to generate its response directly.
- *Image Retrieval and Classification*: We use the original labels directly from the datasets. For classification across meta-categories, we combine subcategories from different meta-categories into a unified training/testing set, where all fine-grained categories are mixed for training.

More specifically, for each task in the human-oriented evaluation, we manually design several question templates to ensure both diversity and comprehensive coverage. More details of curated data can be found in Appendix A.2.

4 Observations and Discussions

4.1 Models under Evaluation

Given the diverse landscape of existing LVLMs, we select nine widely-used open-source LVLMs, two closed-source models (GPT-4o-1120 [1] and Gemini-2.0-flash [13]) and one purely visual model with varying training strategies, as shown in Table 1. To enable clear comparisons and isolate the factors influencing performance, we use earlier versions of models from each family in machine-oriented evaluation, where their distinct characteristics are more evident and less affected by additional tricks or complex modifications. Further details about the evaluated models can be found in Appendix B.

4.2 Human-oriented Evaluation

In the human-oriented evaluation, we assess the ability of LVLMs to understand visual content and utilize domain-specific knowledge through conversational interactions. Specifically, we first evaluate the domain-specific knowledge embedded in LVLMs by measuring their accuracy in identifying object categories across different levels of granularity, as illustrated in Figure 2. Then, we investigate whether LVLMs exhibit knowledge biases when recognizing different fine-grained categories by

Table 1: Training strategies of the open-source evaluated models. “DINOv2" is a purely visual model. “Con" denotes contrastive loss, “Gen" generative loss, “Mat" image-text matching loss, “Rec" reconstruction loss used in BEiT3, and “Dis" distillation loss used in DINOv2.

Model	Vision Encoder	Loss Function					Training Data		
		Con	Gen	Mat	Rec	Dis	$<$ 100M	100M $\sim$ 1B	$>$ 1B
InternVL3-7B [55]	ViT-L	✓	✓	✓		✓			✓
InternVL-Chat-V1.1 [8]	ViT-6B	✓	✓	✓					✓
LLaVA-1.5-7B [24]	ViT-L		✓				✓		
Qwen2.5-VL-7B [3]	ViT-600M	✓	✓	✓		✓			✓
Qwen-VL-Chat [2]	ViT-G		✓						✓
BLIP-2-FLAN-T5-XL [21]	ViT-G	✓	✓	✓				✓	
EVA-CLIP [38]	ViT-L	✓							✓
BEiT3 [42]	ViT-L				✓		✓		
CoCa [50]	ViT-L	✓	✓						✓
DINOv2 [33]	ViT-L	✓				✓		✓	

ranking their accuracy in answering true/false questions for each category, cf. Figure 3. Additionally, we examine the capability of LVLMs to recognize fine-grained attributes, with results summarized in Table 2. Lastly, Table 3 compares the classification accuracy of LVLMs against state-of-the-art fine-grained tailored models on datasets spanning various domains. Based on these observations, we draw the following conclusions.

LVLMs struggle to distinguish excessively fine-grained categories. As shown in Figure 2, taking InternVL3 [55] as an example, its accuracy in answering true/false and multiple-choice questions declines as the granularity becomes finer. At the class level (*e.g.*, *“Is the class of the object in this image an Insecta/Aves?”*), the model achieves 99.76% accuracy on multiple-choice questions and 99.77% on true/false questions.[2] However, as the granularity narrows to the genus level, where negative categories are drawn from different genera within the same class (*e.g.*, *“Is the object in this image an albatross or a gull?”*), the model’s accuracy on multiple-choice questions drops to 90.75%, a decrease of 9.01%. When the granularity is further refined to the species level, where negative categories are from different species within the same genus (*e.g.*, *“Is the object in this image a black-footed albatross/Laysan albatross?”*), the accuracy drops further to 62.48% on true/false questions and 61.18% on multiple-choice questions. This demonstrates that the model struggles to distinguish between closely related species. The results for other LVLMs exhibit similar trends to those observed for InternVL3. Additional examples of multiple-choice and true/false questions can be found in Appendix C.1.

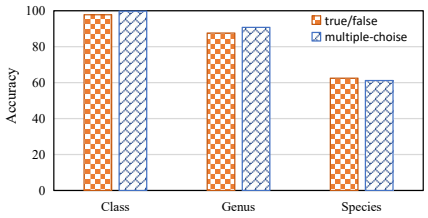

Figure 2: Results of InternVL3 [55] on true/false and multiple-choice questions across different levels of granularity on the *CUB-200-2011* [41] dataset. The x-axis denotes the granularity of the recognition questions.

The inconsistent recognition accuracy of LVLMs across fine-grained categories can be attributed to the characteristics of their training data and the underlying LLM base. To examine whether LVLMs exhibit knowledge bias in recognizing different fine-grained categories, we ranked their accuracy in answering true/false questions for each fine-grained category. As shown in Figure 3, using LLaVA [24] as an example, the model demonstrates highly inconsistent recognition abilities across categories, achieving nearly 90% accuracy for some while dropping to approximately 30% for others.

We hypothesize that this inconsistency arises either from the uneven representation of fine-grained knowledge in the training data or from the inherent difficulty LVLMs face in learning certain fine-grained categories. To test this hypothesis, we fine-tuned LVLMs on datasets with balanced occurrences of fine-grained categories and evaluated their performance. As indicated by the yellow dots in Figure 3, the fine-tuned LLaVA shows consistently strong recognition ability across all fine-grained categories. It suggests that the knowledge bias is primarily caused by the uneven

[2]When questions are relatively simple, LVLMs achieve very high accuracy. The higher accuracy on multiple-choice questions compared to true/false questions may result from randomness.

Table 2: Attribute recognition accuracy of InternVL3 [55] on the *CUB-200-2011* [41] dataset (values in parentheses represent the average accuracy for each attribute).

Color Attribute (47.40)							
belly color	58.49	back color	34.98	bill color	51.31	breast color	54.25
crown color	55.30	eye color	84.59	forehead color	53.32	leg color	44.01
nape color	39.24	throat color	52.77	under tail color	34.69	underparts color	56.20
upper tail color	37.30	upperparts color	28.75	wing color	30.16	primary color	43.05
Pattern Attribute (50.13)							
back pattern	40.94	belly pattern	68.13	breast pattern	65.12	head pattern	35.92
tail pattern	41.64	wing pattern	49.04				
Shape Attribute (30.95)							
bill shape	37.61	shape	52.37	tail shape	10.42	wing shape	23.39
Length Attribute (71.03)				**Size Attribute (52.55)**			
bill length		71.03		size		52.55	

representation of fine-grained knowledge in training data, rather than the inherent difficulty of learning specific categories.

To further validate this hypothesis, we examined the occurrence frequency of fine-grained categories in the LVLM's training data. Interestingly, we found that such categories are almost absent from the training data. This indicates that the observed inconsistency in recognition ability is inherited from the language model underlying the LVLM, rather than being solely a consequence of the visual model itself. Additional results for other LVLMs exhibit similar trends and can be found in Appendix C.2.

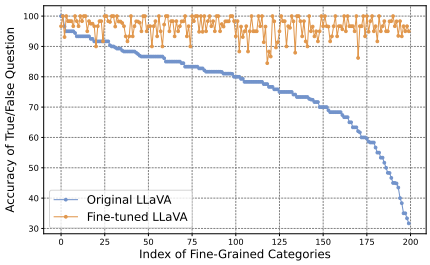

Figure 3: Comparison of the original and fine-tuned LLaVA models on occurrence-balanced fine-grained bird categories. True/false question accuracy for each category is ranked, with blue dots representing the original model and yellow dots the fine-tuned model.

LVLMs exhibit significant room for improvement in recognizing fine-grained attributes. As shown in Table 2 of the paper and Table 12 in the appendix, LVLMs exhibit relatively stronger recognition abilities for attributes like pattern, size and length compared to others. For example, InternVL3 [55] and Qwen2.5-VL [3] achieve 50.13% and 45.12% average accuracy for pattern recognition, but only 30.95% and 29.30% for shape recognition. Notably, only a few attributes achieved an accuracy above 70%, and several attributes scored as low as 10%, highlighting significant room for improvement in LVLMs' fine-grained attribute recognition.

We also observe model-specific differences in attribute recognition. For instance, while InternVL3 struggles more with pattern recognition compared to size, Gemini-2.0-flash [13] shows the opposite trend. Additionally, we find that the latest LVLMs show significant improvements in pattern and length recognition, while their advancements in color and shape recognition are more limited. Detailed results of the attribute recognition task are provided in Appendix C.3.

LVLMs do not outperform fine-grained tailored models in fine-grained tasks. In Table 3, we compare the recognition accuracy of LVLMs with that of fine-grained tailored models. While LVLMs demonstrate notable performance in fine-grained recognition, their accuracy—whether in short-answer questions or using the linear classifier method—falls short of that achieved by fine-grained tailored models. This disparity may stem from the fact that LVLMs are primarily optimized for general tasks, with less emphasis placed on fine-grained domain capabilities. Enhanc-

Table 3: Comparison of LVLMs and fine-grained tailored models on classification tasks. "SA" denotes LVLM results fine-tuned on fine-grained datasets for short-answer questions, "LC" represents linear classifier results using LVLM visual features, and "FG-Tailored" refers to state-of-the-art fine-grained tailored model results.

Datasets	SA	LC	FG-Tailored
CUB-200-2011 [41]	85.60	91.65	93.10 [11]
Stanford Dog [18]	86.49	90.50	97.30 [6]
Stanford Car [19]	90.55	94.30	97.10 [23]
Food-101 [7]	95.25	95.67	98.60 [5]
FGVC Aircraft [29]	66.19	78.88	95.40 [36]

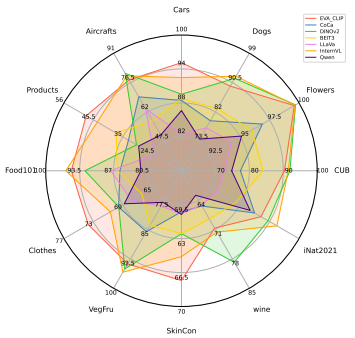

Figure 4: Retrieval results of LVLM visual features on twelve fine-grained datasets. Different colors represent different models.

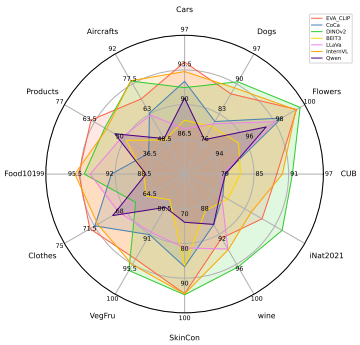

Figure 5: Classification results of LVLM visual features on twelve fine-grained datasets. Different colors represent different models.

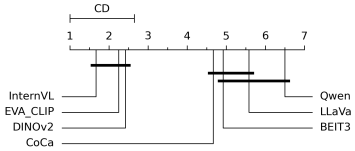

Figure 6: Nemenyi statistical test results for fine-grained retrieval. Black horizontal lines indicate the critical distance (CD), grouping models with no significant performance differences.

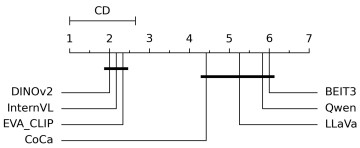

Figure 7: Nemenyi statistical test results for fine-grained recognition. Black horizontal lines indicate the critical distance (CD), grouping models with no significant performance differences.

ing their performance in specialized areas (*e.g.*, fine-grained recognition) while preserving their strengths in general tasks presents an important and promising direction for future research.

4.3 Machine-oriented Evaluation

Compared to human-oriented evaluation, machine-oriented evaluation takes a more direct approach to assess the visual feature representations of LVLMs by employing two fundamental visual tasks: image retrieval and image recognition. This evaluation focuses on two key aspects: 1) Discriminability, which ensures the features can effectively distinguish between fine-grained categories, and 2) Robustness, which examines whether classification accuracy is maintained when visual features are subjected to perturbations.

To evaluate discriminability, we start by analyzing the performance of visual features on retrieval and classification tasks, following the setting of DINOv2 [33]. Figures 4 and 5 display retrieval and classification results across twelve fine-grained datasets, while Figures 6 and 7 provide the Friedman test results for these tasks.

Then, we increase the classification difficulty by merging fine-grained categories from different super-categories into a unified training and testing set. Figure 8 shows the classification accuracy both within a single super-category and across multiple meta-categories. Visual features with strong discriminability should demonstrate high accuracy even when faced with diverse data sources. Furthermore, we examine how the size of the vision encoder in LVLMs affects classification accuracy, as illustrated in Figure 9.

Alignment, a critical component of LVLMs, bridges the gap between vision and language modalities. To explore its influence on visual features, we compare the classification accuracy of original visual features with those aligned to textual features in fine-grained classification tasks, as shown in Table 4.

To investigate robustness, we introduce perturbations to the visual features using projected gradient descent [28] and analyze their impact on classification accuracy. The results of this experiment are presented in Table 5.

The contrastive training paradigm in LVLMs effectively enhances the fine-grained discriminability of visual features. In Figure 4 and Figure 5, vision encoders of LVLMs trained with contrastive paradigms (*e.g.*, EVA-CLIP, InternVL, DINOv2) outperform those trained with reconstruction (BEiT3) and generative paradigms (Qwen) on fine-grained retrieval and classification tasks.

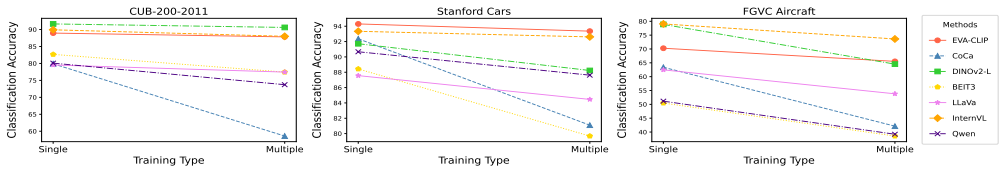

Figure 8: Classification results of LVLM visual features on fine-grained datasets. “Single" denotes accuracy from training on a single meta-category, while “Multiple" reflects accuracy from training on a unified dataset combining multiple meta-categories.

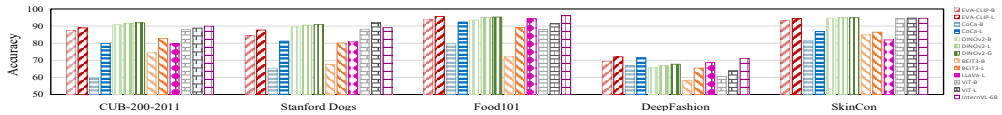

Figure 9: Classification results with different vision encoder sizes. Bars filled with different patterns represent different models, with darker patterns indicating larger vision encoder sizes.

As shown in Figure 6 and Figure 7, the Nemenyi statistical test results also indicate that InternVL, EVA-CLIP and DINOv2 perform significantly better than Qwen and BEiT3. In the multiple meta-category classification results (cf. Figure 8), EVA-CLIP maintains high accuracy, with an average decrease of just 1.96% compared to the single meta-category setting, while Qwen and BEiT3 show larger drops of 4.16% and 7.41%, respectively. This underscores the effectiveness of contrastive training paradigms in fine-grained tasks.

In experiments of vision encoder size, as shown in Figure 9, we observe that DINOv2-B, with a smaller vision encoder, achieves higher classification accuracy compared to BEiT3-L, with a larger vision encoder, outperforming by 8.08% on *CUB-200-2011* and 9.49% on *Stanford Dogs*. We attribute these findings to the limitations of generative and reconstruction training paradigms, which fail to adequately distinguish between features of different fine-grained categories, while maintaining similarity within the same. This limitation hinders their performance on fine-grained tasks.

The alignment strategy in LVLMs might impair the fine-grained discriminability of visual features. Alignment plays a critical role in LVLMs, bridging the gap between visual and textual features. To investigate its impact on fine-grained visual feature discriminability, we compare the classification accuracy of LLaVA’s [24] original visual features with those aligned to textual features on fine-grained classification tasks. As shown in the first two columns of Table 4, the original visual features demonstrate superior classification capabilities, outperforming the aligned features by an average of 3.39%.

Table 4: Classification results of LLaVA visual features on fine-grained datasets before and after alignment with textual features. “Origin" refers to results from the original visual features extracted by the vision encoder. “Aligned" indicates results after alignment with text features using data with inconsistent granularity. “Aligned-FG" shows results after alignment with data of matching visual-text granularity.

Datasets	Origin	Aligned	Aligned-FG
CUB-200-2011	79.77	73.17	75.06
Stanford Dogs	81.24	78.14	80.69
Stanford Cars	87.57	83.90	85.63
Food-101	94.27	93.35	94.32
DeepFashion	69.94	67.30	67.75

This decline in performance can be attributed to two key factors. First, aligning visual and textual features may introduce distortions due to inconsistencies between their respective feature spaces. Second, granularity inconsistencies in LVLMs’ alignment data—where fine-grained objects in images are paired with coarse-grained textual descriptions (as demonstrated in our qualitative analysis in Appendix D.1)—negatively affect the discriminability of the aligned visual features.

To validate the impact of granularity inconsistency, we construct a new alignment dataset where the textual descriptions match the granularity of the fine-grained objects in images. We then retrain the alignment module in LLaVA using this dataset. As shown in Table 4, the classification accuracy of visual features aligned with fine-grained textual content improves significantly, with an increase of 2.55% on *Stanford Dogs* and 1.73% on *Stanford Cars*. While the aligned visual features still underperform compared to the original features, these results underscore the detrimental effects of granularity inconsistency on visual feature discriminability and highlight the importance of ensuring granularity alignment between image and textual content.

Table 5: Classification results of LVLMs' original and perturbed visual features on the fine-grained dataset *CUB-200-2011* and the generic dataset *CIFAR-100*. "Origin" refers to results with original features, while "Perturbed" indicates results with perturbed features.

Datasets	EVA-CLIP		CoCa		DINOv2		ViT	
	Origin	Perturbed	Origin	Perturbed	Origin	Perturbed	Origin	Perturbed
CIFAR-100	93.05	50.76	86.94	52.23	93.38	42.39	89.81	72.15
CUB-200-2011	88.95	24.94	79.89	23.40	91.64	25.94	88.83	73.85

Larger vision encoder size or larger scale of web data provides limited improvement in the fine-grained discriminability of visual features. Regarding vision encoder size, as shown in Figure 9, increasing the size of DINOv2's vision encoder from DINOv2-B to DINOv2-L results in only a 0.6% improvement in average classification accuracy, while further increasing from DINOv2-L to DINOv2-G leads to just a 0.3% improvement. The classification accuracy of visual features in InternVL-6B is not superior to that of DINOv2-L, suggesting that merely enlarging the vision encoder has limited impact on enhancing the fine-grained discriminability of visual features.

Regarding the amount of training data, as shown in Figure 7, the vision encoder of EVA-CLIP, trained on over 2 billion data samples, does not outperform DINOv2, which was trained on 142 million samples, in fine-grained classification and retrieval tasks. We attribute this difference to the quality of the training data, that DINOv2's dataset is carefully curated from a large pool of data, whereas EVA-CLIP relies on crawled web data. A similar trend is observed when comparing DINOv2 with InternVL (6B samples), suggesting that simply increasing the scale of training data, without considering its quality, offers limited improvement in fine-grained discriminability of visual features.

Visual features in LVLMs are more susceptible to perturbations in fine-grained tasks. As shown in Table 5, perturbing the visual features of EVA-CLIP causes a significant drop in classification accuracy on the fine-grained dataset *CUB-200-2011* (from 88.95% to 24.94%). In contrast, on the generic dataset *CIFAR-100* [20], the accuracy declines more modestly (from 93.05% to 50.76%). Similar trends are observed for CoCa and DINOv2, indicating that fine-grained tasks are more susceptible to perturbations than generic tasks. We attribute this vulnerability to the coarse-grained and noisy data used during training, which limits the discriminability of features across fine-grained categories, making it challenging to distinguish between them when visual features are perturbed.

In contrast, the Vision Transformer [46] trained on the *ImageNet* [10] dataset with cross-entropy loss demonstrates superior robustness to perturbations. It shows only minor declines in classification accuracy on both fine-grained and generic datasets. This suggests that adopting alternative training paradigms or incorporating high-quality, fine-grained data (as seen in *ImageNet*) during training could help improve the robustness of visual features in LVLMs.

5 Concluding Remarks

We conducted a comprehensive evaluation to investigate the capabilities of LVLMs on fine-grained visual tasks, leading to several key findings. First, we observed that the contrastive training paradigm significantly enhances the fine-grained discriminability of visual features, whereas generative and reconstruction-based paradigms tend to underperform in this aspect. Second, aligning visual features with textual features can impair their fine-grained discriminability, particularly when there is a mismatch in granularity between the paired image-text data. Third, our experiments revealed that LVLMs and LVMs are more susceptible to feature perturbations in fine-grained tasks compared to generic vision tasks, highlighting a vulnerability that merits further attention. Fourth, while LVLMs demonstrate strong capabilities in visual appearance perception, they face notable challenges in fine-grained category reasoning, which depends heavily on recognizing subtle visual attributes. Finally, despite their advancements, LVLMs still lag behind specialized fine-grained models, underscoring the need for task-specific enhancements. These findings highlight key limitations of LVLMs, identifying critical areas for improvement, such as training paradigms, robustness to perturbations, and reasoning capabilities. Our work lays the groundwork for future research to advance LVLM performance on fine-grained visual tasks.

Limitations. Despite `FG-BMK` covers most fine-grained image tasks, it is still impossible to cover *every* fine-grained scenario in real-world applications, such as medical imaging or remote sensing.

Nevertheless, we believe that FG-BMK provides the most practical setting for evaluating LVLMs in most use cases, and the conclusions drawn from this benchmark should be applicable to real-world fine-grained visual tasks.

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A The Evaluation Benchmark

A.1 Evaluation Task Details

In Section 3.1, we have described each evaluation task. Here, we provide further details. In the Knowledge Bias Estimation task, to uncover potential knowledge biases across different fine-grained categories, we pair each image with its corresponding fine-grained label to generate positive samples for true/false questions. For constructing negative samples, each image is paired with a single fine-grained label randomly selected from other subcategories within the same super-category. For each fine-grained category, we calculate the LVLM's accuracy on all coresponding true/false questions as a measure of its understanding of that category's knowledge.

In the cross meta-class classification task, we follow the DINOv2 [33] method to train the model on a unified training set where fine-grained categories from different datasets are combined. The model is then tested on each individual dataset to evaluate its performance.

A.2 Data Curation

Dataset We source images for the `FG-BMK` benchmark from 12 fine-grained datasets. These datasets cover a wide range of meta-classes, with different categories and sample, providing a comprehensive assessment of LVLMs capabilities on fine-grained tasks across different domains. Table 6 indicates their meta-classes, the amount of samples, the number of categories. For all datasets, we construct human-oriented evaluation questions based on their test sets. We use the original labels directly from the datasets for the machine-oriented evaluation.

Table 6: Details of 12 fine-grained datasets sorted by their numbers of categories. "Meta-class" refers to a high-level categorization of the dataset. "Categories" refers to the number of fine-grained categories. "Samples" refers to the total number of samples in each dataset.

Datasets	Meta-class	Categories	Samples
Wine [39]	Industrial	11	4,516
DeepFashion [26]	Clothes	46	18,000
SkinCon [9]	Dermatology	48	3,866
Flowers102 [32]	Flower	102	7,169
Food101 [7]	Food	101	101,000
FGVC Aircraft [29]	Aircraft	100	6,667
Stanford Dogs [18]	Dog	120	20,580
Stanford Cars [19]	Car	196	16,185
CUB-200-2011 [41]	Bird	200	11,788
VegFru [14]	Vegetable	292	146,131
Products-10K [4]	Retail	9,691	197,307
iNat2021 [40]	Plants	10,000	2,786,843

Human-oriented Question Templates When constructing true/false, multiple-choice, short answer questions for each task in human-oriented evaluation, we manually design several question templates to ensure both diversity and comprehensive coverage. Figure 10 illustrates the question templates we use for generating the tasks.

B Evaluated Models

As shown in Table 7, we select nine widely-used open-source LVLMs, two closed-source models (GPT-4o [1] and Gemini-2.0-flash [13]) and one purely visual model, each of which employs a distinctive training recipes, including variations in vision encoder, language model, training losses and data.

- **EVA-CLIP** [38] aligns visual and textual features using contrastive loss, leveraging over 2 billion web image-text pairs and advanced optimization techniques.

Question Template for Attribute Recognition Task

True/False Question:
Is the wing color of the bird {color}?
Is the breast pattern of the bird {pattern}?

Multiple-choice Question:
What is the wing color of the bird? Choose one from the following list: {options}.

Question Template for Knowledge Bias Estimation Task

True/False Question:
Is the species of the {meta_class} {fine_grained_category}? Answer with yes or no.
Does this {meta_class} belong to the species known as {fine_grained_category}? Answer with yes or no.
Is the {meta_class} species in this photo a {fine_grained_category}? Answer with yes or no.
From your observation, is the species of the {meta_class} shown a {fine_grained_category}? Answer with yes or no

Question Template for Hierarchical Granularity Recognition Task

True/False Question:
Is the species of the {meta_class} {fine_grained_category}? Answer with yes or no.
Does this {meta_class} belong to the species known as {fine_grained_category}? Answer with yes or no.
Is the {meta_class} species in this photo a {fine_grained_category}? Answer with yes or no.
From your observation, is the species of the {meta_class} shown a {fine_grained_category}? Answer with yes or no

Multiple-choice Question:
What is the species of the {meta_class} in this image? Choose one answer from the following list: {options}. Answer the question using a single word or phrase.
Can you identify the species of this {meta_class}? Choose one answer from the following list: {options}. Answer the question using a single word or phrase.
Which species does this {meta_class} in the photo belong to? Choose one answer from the following list: {options}. Answer the question using a single word or phrase.
Observing the {meta_class} in the image, which of the following species is it? Choose one answer from the following list: {options}. Answer the question using a single word or phrase.

Short Answer question:
What species of the {meta_class} is shown in the image? Directly answer with species names.
Can you identify the species of this {meta_class} from the image? Directly answer with species names.
Which species does the {meta_class} in this photo belong to? Directly answer with species names.
Based on your observation, what species of the {meta_class} is depicted? Directly answer with species names.

Figure 10: Question templates for each task in huamn-oriented evaluation.

- **InternVL3** [55] adopts a unified pre-training approach over both multimodal and pure-text data, enhanced by variable visual position encoding (V2PE) and advanced post-training strategies for improved scalability and effectiveness.
- **InternVL** [8] leverages contrastive, matching, and generative losses in a multi-stage training process, with a large-scale vision encoder and over 6 billion image-text pairs to align visual and textual representation.
- **BLIP-2** [21] bridges the modality gap between frozen image encoders and LLMs using a lightweight Q-Former, leveraging contrastive, matching, and generative loss in a two-stage pre-training process over 129 million data with fewer trainable parameters.
- **Qwen2.5-VL** [3] combines dynamic-resolution Vision Transformer with Window Attention to reduce computational cost while preserving native image resolution.
- **Qwen-VL** [2] employs a three-stage training process with generative loss, using a VL adapter to align visual and textual features while reducing computational cost over 1.4 billion image-text pairs.

Table 7: Configurations of the evaluated models. “DINOv2-L” is a purely visual model. “Con” stands for the contrastive loss, “Gen” for the generative loss, “Mat” for the image-text matching loss, “Rec” for the reconstruction loss as used in BEiT3 [42], and “Dis” for the distillation loss as applied in DINOv2 [33].

Model	Component		Loss Function				
	Vision Model	Language Model	Con	Gen	Mat	Rec	Dis
InternV3-7B	InternViT-L	Qwen2.5-7B	✓	✓	✓		✓
InternVL-Chat-V1.1	InternViT-6B	LLaMA2-13B	✓	✓	✓		
LLaVA-1.5-7B	CLIP-L	Vicuna-7B		✓			
Qwen2.5-VL	CLIP-600M	Qwen2.5-7B	✓	✓	✓		✓
Qwen-VL	Openclip-G	Qwen-7B		✓			
BLIP-2-FLAN-T5-XL	EVA-CLIP-G	FlanT5-XL	✓	✓	✓		
EVA02-CLIP-L	EVA02-L	CLIP-L	✓				
BEiT3-L-ITC	CLIP-L	CLIP-L				✓	
CoCa-L	CLIP-L	CLIP-L	✓	✓			
DINOv2-L	CLIP-L	/	✓				✓

- **CoCa** [50] adopts task-specific attentional pooling to tailor visual representations for different training objectives, applying contrastive loss to train the first half of the decoder and generative loss to train the full decoder in an end-to-end manner over 5 billion image-text pairs.
- **BEIT3** [42] treats images as a foreign language, leveraging a mask-then-predict objective over 36 million image-text pairs to unify vision and language pretraining, and introduces a multiway transformer architecture for general-purpose modeling.
- **LLaVA** [24] aligns visual and textual features using a simple MLP with generative loss, leveraging 1.2 million GPT-4 [] generated multimodal instruction-following data for training.
- **DINOv2** [33] uses a self-supervised learning approach, leveraging knowledge distillation and a mask-then-predict strategy over 142 million images to train the vision encoder.

For all our evaluated model, we follow their official configurations to run the inference. We set the temperature of all open-source models to 0, while keeping the default for closed-source APIs.

C Human-oriented Evaluations

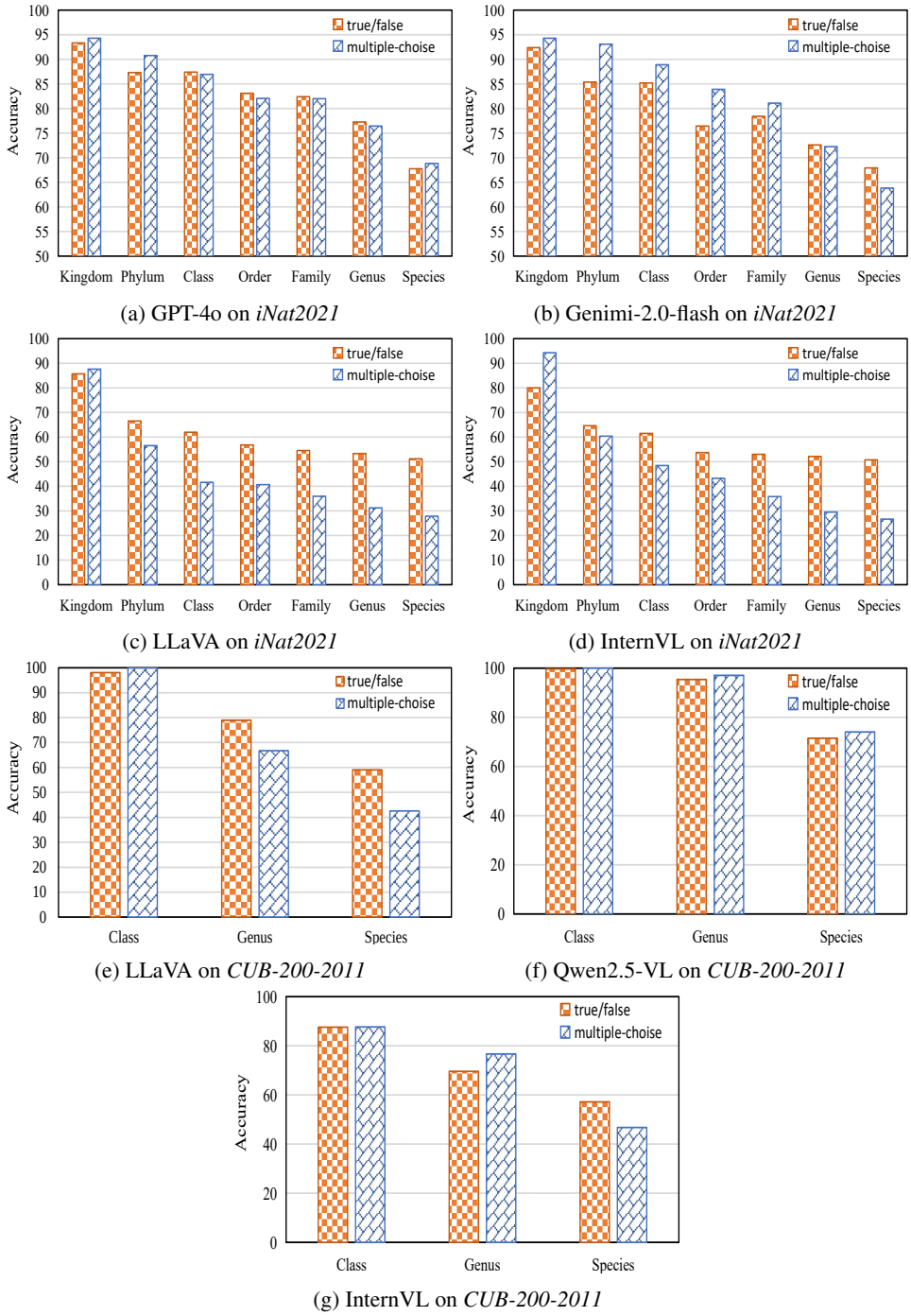
C.1 Results of Hierarchical Granularity Recognition

Figure 2 shows InternVL3’s [55] accuracy in answering true/false and multiple-choice questions within hierarchical granularity recognition task on *CUB-200-2011* dataset. In Figure 11, we present additional results for GPT-4o [1], Gemini-2.0-flash [13], Qwen2.5-VL [3], LLaVA [24] and InternVL [8] on *CUB-200-2011* [41] and *iNat2021* [40] datasets. As shown in the experiments, the accuracy of all models decreases as the granularity becomes finer. When the granularity level reaches the finest level, the models struggle to distinguish between closely related species.

C.2 Results of Knowledge Bias Estimation

In Figure 3, we observe that LLaVA exhibit highly inconsistent recognition abilities across categories. We also conduct experiments with Qwen2.5-VL, GPT-4o, and Gemini-2.0-flash on fine-grained datasets such as Aircraft [29], Flowers102 [32] and Stanford Dogs [18]. As shown in Figure 12 and Figure 13, all LVLMs display similar trends, indicating inconsistent recognition abilities across fine-grained categories. However, after fine-tuned on datasets with balanced occurrences of fine-grained categories, LVLMs demonstrate remarkable recognition abilities across all fine-grained categories.

To construct datasets with balanced occurrences of fine-grained categories, we select an equal number of images from each category. Then we generate the same number of true/false questions for each fine-grained category, thereby fine-tuning the LVLMs in a way that each category receives balanced representation.



(a) GPT-4o on *iNat2021*

(b) Genimi-2.0-flash on *iNat2021*

(c) LLaVA on *iNat2021*

(d) InternVL on *iNat2021*

(e) LLaVA on *CUB-200-2011*

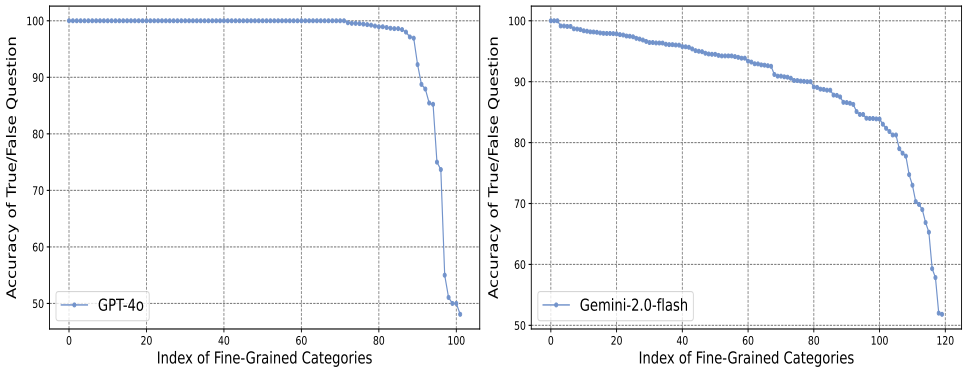
(f) Qwen2.5-VL on *CUB-200-2011*

(g) InternVL on *CUB-200-2011*

Figure 11: Results of GPT-4o [1], Gemini-2.0-flash [13], Qwen2.5-VL [3], LLaVA [24] and InternVL [8] on true/false and multiple-choice questions across different levels of granularity on *CUB-200-2011* [41] and *iNat2021* [40] dataset. The x-axis denotes the granularity of the recognition questions.

C.3 Results of Attribute Recognition

Table 2 and Table 12 shows the attribute recognition accuracy of InternVL3 and Qwen2.5-VL on the *CUB-200-2011* dataset. The results of LLaVA, BLIP2, InternVL and Gemini-2.0-flash are shown in Table 8, Table 9, Table 10, and Table 11.



(a) GPT-4o on *Flowers102*

(b) Gemini-2.0-flash on *Stanford Dog*

Figure 12: Knowledge bias estimation results of two closed-source models. True/false question accuracy for each category is ranked, with blue dots representing the original model.

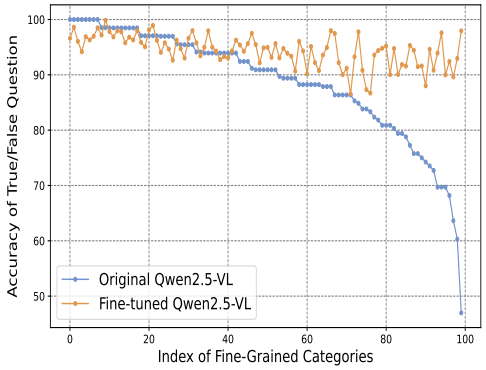


Figure 13: Comparison of the original and fine-tuned Qwen2.5-VL [3] models on occurrence-balanced fine-grained aircraft categories. True/false question accuracy for each category is ranked, with blue dots representing the original model and yellow dots the fine-tuned model.

Table 8: Attribute recognition accuracy of LLaVA [24] on the *CUB-200-2011* [41] dataset (values in parentheses represent the average accuracy for each attribute).

Color Attribute (44.34)							
belly color	54.79	back color	41.90	bill color	41.44	breast color	49.56
crown color	48.71	eye color	69.27	forehead color	47.03	leg color	35.37
nape color	40.51	throat color	35.40	under tail color	38.88	underparts color	54.81
upper tail color	41.41	upperparts color	34.00	wing color	34.60	primary color	41.77
Pattern Attribute (23.69)							
back pattern	27.27	belly pattern	26.41	breast pattern	24.24	head pattern	11.35
tail pattern	23.19	wing pattern	29.67				
Shape Attribute (14.05)							
bill shape	1.39	shape	18.59	tail shape	9.89	wing shape	26.34
Length Attribute (15.71)				**Size Attribute (49.47)**			
bill length		15.71		size		49.47	

Table 9: Attribute recognition accuracy of BLIP2 [21] on the *CUB-200-2011* [41] dataset (values in parentheses represent the average accuracy for each attribute).

Color Attribute (37.94)							
belly color	51.15	back color	39.64	bill color	23.42	breast color	50.17
crown color	42.59	eye color	23.59	forehead color	43.18	leg color	18.77
nape color	41.55	throat color	53.81	under tail color	37.98	underparts color	41.60
upper tail color	37.52	upperparts color	33.01	wing color	31.25	primary color	33.48
Pattern Attribute (11.34)							
back pattern	14.66	belly pattern	7.82	breast pattern	9.48	head pattern	2.14
tail pattern	14.21	wing pattern	19.73				
Shape Attribute (25.05)							
bill shape	8.84	shape	34.69	tail shape	13.51	wing shape	43.19
Length Attribute (30.11)				**Size Attribute (27.62)**			
bill length		30.11		size		27.62	

Table 10: Attribute recognition accuracy of InternVL [8] on the *CUB-200-2011* [41] dataset (values in parentheses represent the average accuracy for each attribute).

Color Attribute (35.78)							
belly color	52.09	back color	33.89	bill color	26.59	breast color	46.58
crown color	39.91	eye color	23.68	forehead color	40.83	leg color	32.75
nape color	29.66	throat color	30.21	under tail color	32.31	underparts color	50.57
upper tail color	33.42	upperparts color	29.64	wing color	27.17	primary color	40.15
Pattern Attribute (34.71)							
back pattern	35.57	belly pattern	44.14	breast pattern	42.22	head pattern	11.81
tail pattern	35.86	wing pattern	37.31				
Shape Attribute (23.03)							
bill shape	12.16	shape	38.08	tail shape	15.49	wing shape	26.43
Length Attribute (29.31)				**Size Attribute (47.70)**			
bill length		29.31		size		47.70	

Table 11: Attribute recognition accuracy of Gemini-2.0-flash [13] on the *CUB-200-2011* [41] dataset (values in parentheses represent the average accuracy for each attribute).

Color Attribute (47.22)							
belly color	62.09	back color	36.51	bill color	52.31	breast color	56.01
crown color	56.44	eye color	59.57	forehead color	53.55	leg color	40.66
nape color	40.40	throat color	60.23	under tail color	40.60	underparts color	59.65
upper tail color	39.99	upperparts color	29.66	wing color	29.21	primary color	38.69
Pattern Attribute (56.14)							
back pattern	56.26	belly pattern	70.51	breast pattern	66.89	head pattern	39.56
tail pattern	52.33	wing pattern	51.26				
Shape Attribute (48.75)							
bill shape	61.62	shape	68.20	tail shape	32.13	wing shape	33.04
Length Attribute (71.82)				**Size Attribute (52.72)**			
bill length		71.82		size		52.72	

Table 12: Attribute recognition accuracy of Qwen2.5-VL [3] on the *CUB-200-2011* [41] dataset (values in parentheses represent the average accuracy for each attribute).

Color Attribute (40.39)							
belly color	51.11	back color	32.89	bill color	46.50	breast color	44.84
crown color	46.54	eye color	54.85	forehead color	44.57	leg color	37.79
nape color	36.49	throat color	40.74	under tail color	34.60	underparts color	50.20
upper tail color	34.92	upperparts color	27.20	wing color	26.03	primary color	36.96
Pattern Attribute (45.12)							
back pattern	42.66	belly pattern	64.58	breast pattern	59.79	head pattern	14.57
tail pattern	45.04	wing pattern	44.11				
Shape Attribute (29.30)							
bill shape	15.30	shape	58.17	tail shape	5.63	wing shape	38.10
Length Attribute (63.20)				**Size Attribute (52.56)**			
bill length		63.20		size		52.56	

D Machine-oriented Evaluations

D.1 Qualitative Analysis of Granularity Inconsistency in LVLM Alignment Data

In the LVLM's alignment data, we observe a phenomenon of granularity inconsistency, where fine-grained objects in images are paired with coarse-grained textual descriptions. Figure 14 shows some examples of granularity inconsistency, as well as a constructed sample of properly aligned granularity.

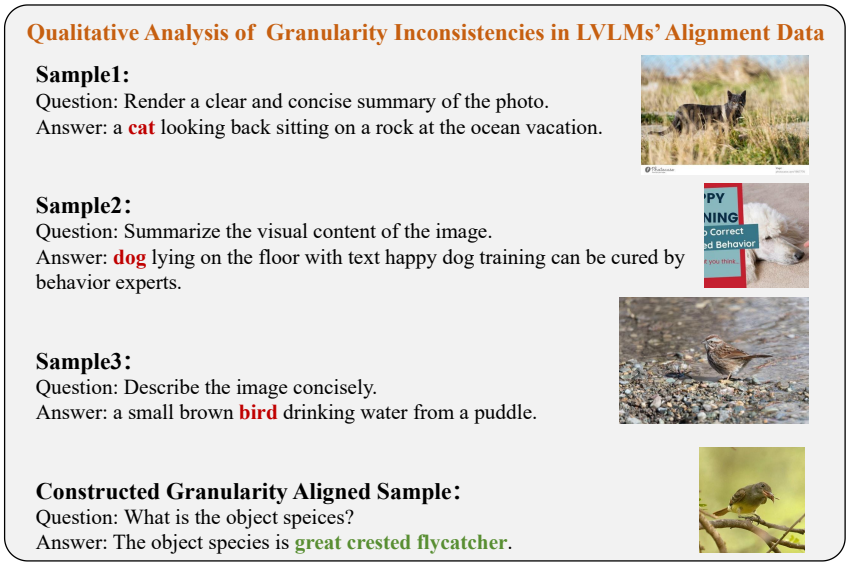


Figure 14: Qualitative analysis of granularity inconsistencies in LVLMs' alignment data and a constructed sample of properly aligned granularity.

D.2 Results of Classification Across Multi-categories

In Figure 8, we have shown the classification accuracy both within a single super-category and across multiple meta-categories in three datasets. Here, in Figure 15, we include more results on nine fine-grained datasets. As shown in the results, EVA-CLIP, trained with contrastive paradigm, maintains a higher score in classification across multiple meta-categories compared to Qwen and BEiT3, which are trained with generative and reconstruction paradigms.

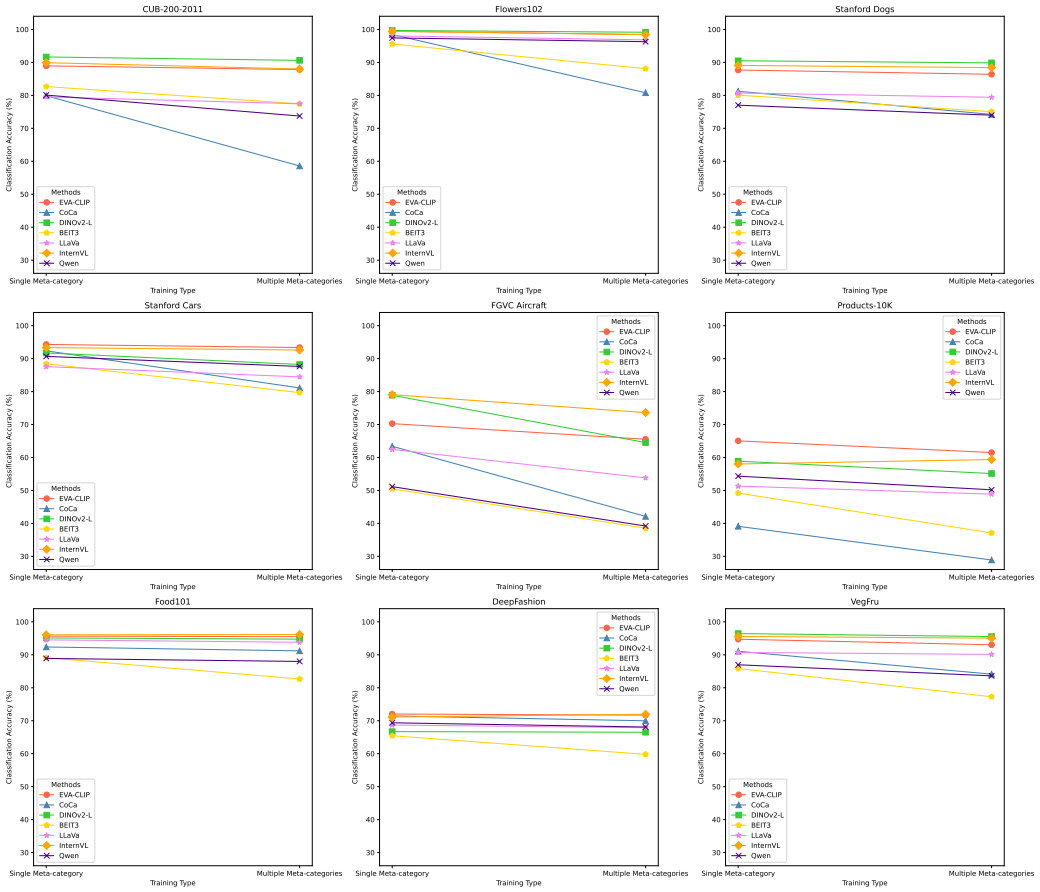


Figure 15: Classification results of LVLM visual features on fine-grained datasets. “Single" denotes accuracy from training on a single meta-category, while “Multiple" reflects accuracy from training on a unified dataset combining multiple meta-categories.